

Кросс-компиляция

1. Сборка библиотеки zlib

Конфигурационные настройки:

```
./configure --prefix=$PWD/_install --static
```

Сборка:

```
CC=arm-linux-gnueabi-gcc make
```

Результат:



```
tair@tair-Bravo-15-C7VE:~/dev/kernel-arm-lesson-ssh/zlib/_install$ ls
include lib share
tair@tair-Bravo-15-C7VE:~/dev/kernel-arm-lesson-ssh/zlib/_install$
```

В компьютер есть два типа часов: аппаратные (продолжают идти даже в выключенном компьютере) и программные (находятся под контролем...

2. Сборка библиотеки openssl

Конфигурационные настройки:

```
./Configure linux-armv4 no-shared no-module --cross-compile-prefix=arm-  
linux-gnueabi- --prefix=$PWD/_install
```

Сборка:

```
make -j6
```

Установка:

```
make install
```

Результат:



```
tair@tair-Bravo-15-C7VE:~/dev/kernel-arm-lesson-ssh/openssl-3.5.7$ cd _install/
tair@tair-Bravo-15-C7VE:~/dev/kernel-arm-lesson-ssh/openssl-3.5.7/_install$ ls
bin include lib share ssl
tair@tair-Bravo-15-C7VE:~/dev/kernel-arm-lesson-ssh/openssl-3.5.7/_install$
```

Постоянно сбывается время на экране, после того как...

3. Сборка библиотеки openssh

Конфигурационные настройки:

```
./configure --host=arm-linux-gnueabi --with-zlib=$PWD/../zlib/_install --  
with-ssl-dir=$PWD/../openssl-3.5.7/_install --prefix=$PWD/_install --  
disable-strip LDFLAGS="-static"
```

Сборка:

make -j6

Установка:

make install

Результат:

```
tair@tair-Bravo-15-C7VE:~/dev/kernel-arm-lesson-ssh/openssh-9.9p2$ cd _install/
tair@tair-Bravo-15-C7VE:~/dev/kernel-arm-lesson-ssh/openssh-9.9p2/_install$ ls
bin etc libexec sbin share
tair@tair-Bravo-15-C7VE:~/dev/kernel-arm-lesson-ssh/openssh-9.9p2/_install$ cd bin/
tair@tair-Bravo-15-C7VE:~/dev/kernel-arm-lesson-ssh/openssh-9.9p2/_install/bin$ ls
scp sftp ssh ssh-add ssh-agent ssh-keygen ssh-keyscan
tair@tair-Bravo-15-C7VE:~/dev/kernel-arm-lesson-ssh/openssh-9.9p2/_install/bin$
```

4. Создание пользователя

```
tair@tair-Bravo-15-C7VE:~/dev/kernel-arm-lesson-ssh/_install$ ls
bin dev etc home init libexec linuxrc sbin usr
tair@tair-Bravo-15-C7VE:~/dev/kernel-arm-lesson-ssh/_install$ cd etc/
tair@tair-Bravo-15-C7VE:~/dev/kernel-arm-lesson-ssh/_install/etc$ ls
group inittab passwd shadow
tair@tair-Bravo-15-C7VE:~/dev/kernel-arm-lesson-ssh/_install/etc$
```

5. Файл init

```
#!/bin/ash

# Монтируем системные директории
mount -t devtmpfs devtmpfs /dev

su tair

echo "=====
echo " Welcome Tair. Lesson SSH"

exec /bin/ash
```

6. Архивирование

```
find . -print0 | cpio --null -o -H newc | gzip -9 > ../initramfs.cpio.gz
```

7. Запуск qemu и проверка ssh

```
EMU_AUDIO_DRV=none qemu-system-arm -M vexpress-a9 -kernel  
zImage -initrd initramfs.cpio.gz -dtb vexpress-v2p-ca9.dtb -append  
"console=ttyAMA0" -nographic
```

```
tair@tair-Bravo-15-C7VE: ~/dev/kernel-arm-lesson-ssh  
tair@tair-Bravo-15-C7VE: ~/dev/kernel-arm-lesson-ssh 190x43  
[ 2.569287] usbbid: USB HID core driver  
[ 2.533049] hw perfevents: enabled with armv7_cortex_a9 PMU driver, 7 (0000003f) counters available  
[ 2.535821] NET: Registered PF_INET6 protocol family  
[ 2.541829] Segment Routing with IPv6  
[ 2.541984] In-situ OAM (IOAM) with IPv6  
[ 2.542307] sit: IPv6, IPv4 and MPLS over IPv4 tunneling driver  
[ 2.544342] NET: Registered PF_PACKET protocol family  
[ 2.544375] can: controller area network core  
[ 2.544624] NET: Registered PF_CAN protocol family  
[ 2.544667] can: raw protocol  
[ 2.544775] can: broadcast manager protocol  
[ 2.544878] can: netlink gateway - max_hops=1  
[ 2.546006] Key type dns_resolver registered.  
[ 2.546426] ThumbEE CPU extension supported.  
[ 2.546478] Registering SWP/SWPB emulation handler  
[ 2.608858] Loading compiled-in X.509 certificates  
[ 2.621890] 10009000.serial: ttyAMA0 at MMIO 0x10009000 (irq = 37, base_baud = 0) is a PL011 rev1  
[ 2.623566] printk: console [ttyAMA0] enabled  
[ 2.688218] 1000a000.serial: ttyAMA1 at MMIO 0x1000a000 (irq = 38, base_baud = 0) is a PL011 rev1  
[ 2.704672] 1000b000.serial: ttyAMA2 at MMIO 0x1000b000 (irq = 39, base_baud = 0) is a PL011 rev1  
[ 2.707197] 1000c000.serial: ttyAMA3 at MMIO 0x1000c000 (irq = 40, base_baud = 0) is a PL011 rev1  
[ 2.913253] input: InEXPS/2 Generic Explorer Mouse as /devices/platform/bus@40000000/bus@40000000:motherboard-bus@40000000/bus@40000000:iofpga07,00000000/10007000.kni/serio1/input/input1  
[ 3.030523] input: AT Raw Set 2 keyboard as /devices/platform/bus@40000000/bus@40000000:motherboard-bus@40000000/bus@40000000:iofpga07,00000000/10006000.kni/serio0/input/input2  
[ 3.399346] clk: Disabling unused clocks  
[ 3.399600] PM: genpd: Disabling unused power domains  
[ 3.437613] Freeing unused kernel image (initmem) memory: 2048K  
[ 3.441346] Run /init as init process  
sh: can't access tty: job control turned off  
/ $ ssh  
[ 13.987950] amba 1000f000.wdt: deferred probe pending: (reason unknown)  
[ 13.988022] amba 1000e000.memory-controller: deferred probe pending: (reason unknown)  
[ 13.988032] amba 1000e1000.memory-controller: deferred probe pending: (reason unknown)  
[ 13.988040] amba 1000e5000.watchdog: deferred probe pending: (reason unknown)  
[ 16.827420] random: crng init done  
usage: ssh [-46AaCfGgKkMnNqstTvVXxy] [-B bind_interface] [-b bind_address]  
[-c cipher_spec] [-D [bind_address]:port] [-E log_file]  
[-e escape_char] [-F configfile] [-I pkey1] [-i identity_file]  
[-J destination] [-L address] [-l login_name] [-m mac_spec]  
[-O ctl_cmd] [-o option] [-P tag] [-p port] [-R address]  
[-S ctl_path] [-W host:port] [-w local_tun[:remote_tun]]  
destination [command [argument ...]]  
ssh [-Q query_option]  
/ $
```